Automated qPCR Analysis Report

Experiment: Auto qPCR run **Date:** 2025-11-21 **Analyst:** Auto-analyst

Summary of Results

- TargetX | Control_1: Fold-change 1.11
- TargetX | Control_2: Fold-change 0.90
- TargetX | ExpA_1: Fold-change 11.71
- TargetX | ExpA_2: Fold-change 10.93

QC Summary

Test	Status	Notes
NTC	pass	No amplification in NTC
Replicate variance	Warn	Control_1
Housekeeping	Warn	GAPDH: fail; ACTB: missing

Cleaned CT Table

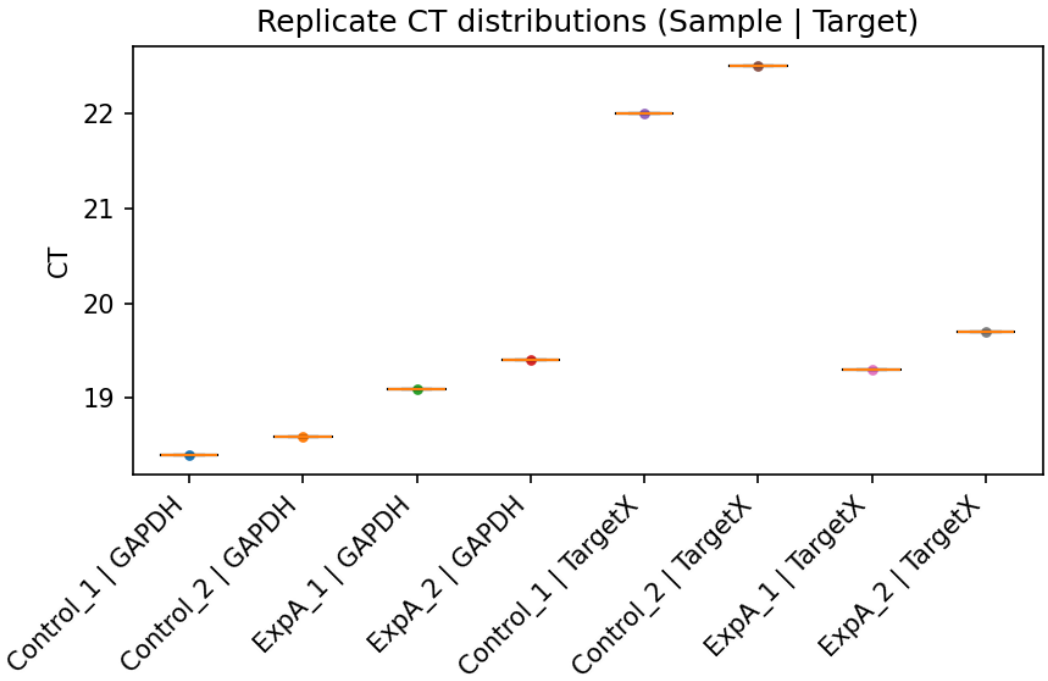
Sample	Target	CT_mean	n_rep	CT_sd
Control_1	GAPDH	18.4	1	nan
Control_1	TargetX	22	1	nan
Control_2	GAPDH	18.6	1	nan
Control_2	TargetX	22.5	1	nan
ExpA_1	GAPDH	19.1	1	nan
ExpA_1	TargetX	19.3	1	nan
ExpA_2	GAPDH	19.4	1	nan
ExpA_2	TargetX	19.7	1	nan
NTC	TargetX	nan	0	nan

ΔCT / ΔΔCT Analysis

Sample	Target	DeltaCT	DeltaDeltaCT	FoldChange
Control_1	TargetX	3.6	-0.15	1.10957
Control_2	TargetX	3.9	0.15	0.90125
ExpA_1	TargetX	0.2	-3.55	11.7127
ExpA_2	TargetX	0.3	-3.45	10.9283

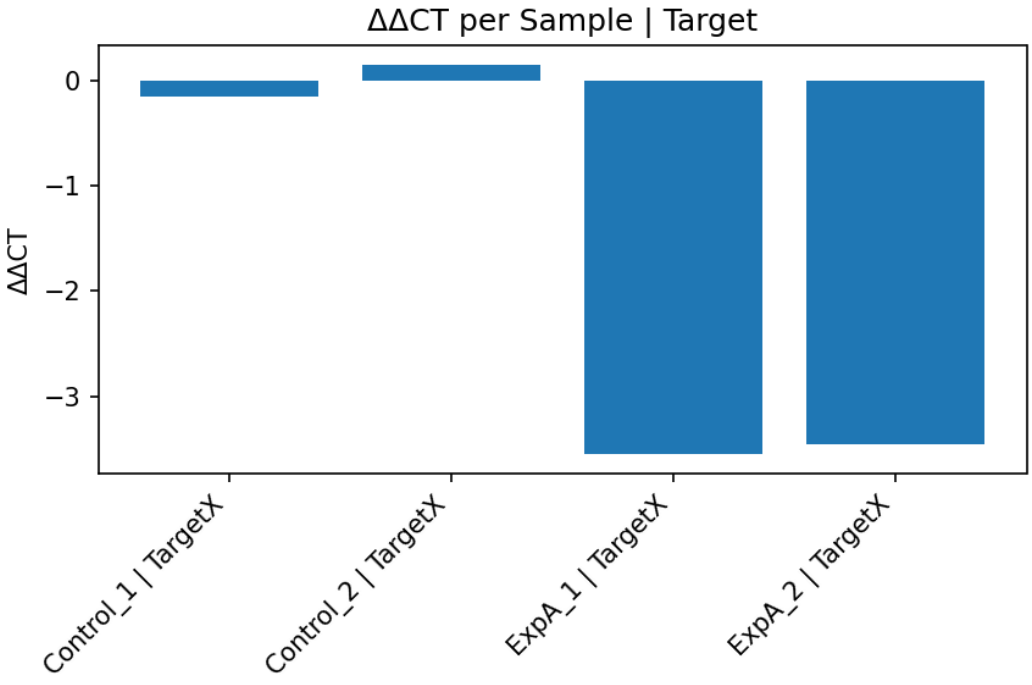
Plots

Replicate distributions



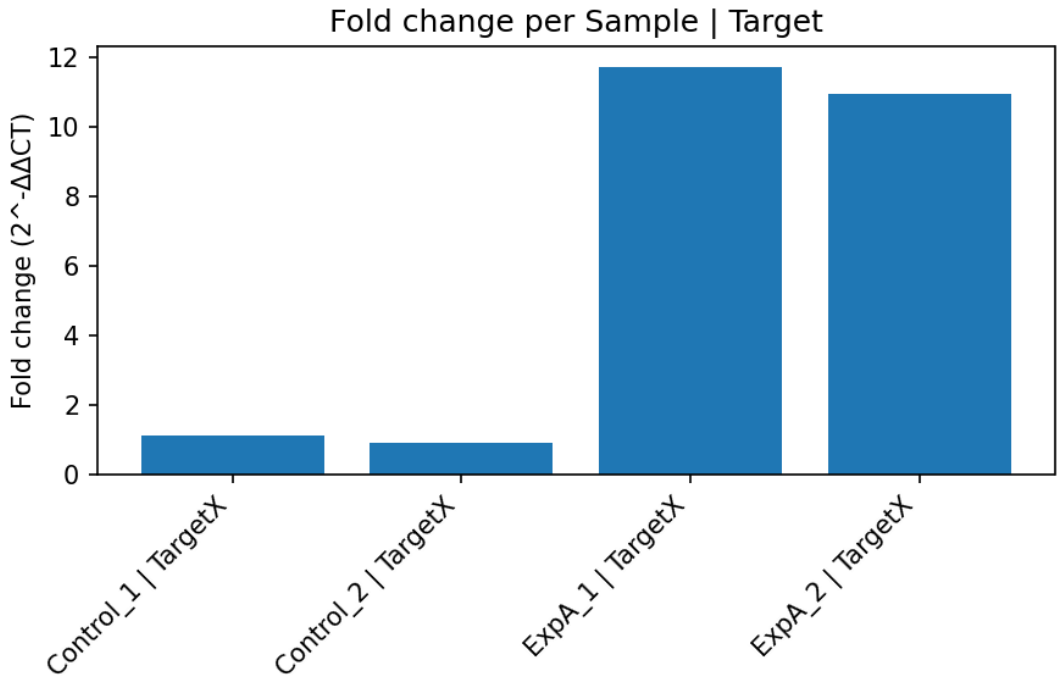
Replicate CT distributions (Sample | Target)

ΔΔCT / Fold-change



ΔΔCT values per Sample | Target

Fold change (2^-ΔΔCT)



Fold change per Sample | Target

Interpretation

The no-template control (NTC) showed no detectable amplification across targets, indicating no evidence of contamination or widespread primer-dimer artifacts.

Housekeeping QC flags detected (failed: GAPDH; missing: ACTB). Normalization may be unreliable for some comparisons.

Technical replicates showed acceptable variance for most samples/targets.

For **TargetX**, experimental samples show mean ΔΔCT = -3.50, corresponding to a mean fold change ≈ 11.32. This suggests the gene is strongly upregulated.

At least one QC criterion failed or raised warnings; interpret fold-change values with caution and consider repeating critical samples.

Generated by automated_qPCR pipeline